Regression algorithm assignment report

1. Problem Statement

The main aim is to predict insurance charges based on multiple user attributes such as age, BMI, number of children, gender, and smoking status. We are required to build regression models that can accurately estimate insurance costs using historical data.

2. Dataset Info

The dataset used for this project is ***insurance_pre.csv***.

Basic Information:

- Total Rows: 1338
- Total Columns: 7
- After preprocessing: 5 input features + 1 target variable

Features:

- age (Numerical)
- sex (Categorical)
- bmi (Numerical)
- children (Numerical)
- smoker (Categorical)
- region (Categorical - removed/encoded)

Target Variable:

- charges (Insurance cost)

3. Data Preprocessing

The following preprocessing steps were performed:

- Categorical variables were converted into numerical format using:
  - ***pd.get_dummies()***
- One-hot encoding was applied with ***drop_first=True*** to avoid dummy variable trap.
- Features selected:
  - age
  - sex_male
  - bmi
  - children
  - smoker_yes

4. ML Models Used

The following regression algorithms were applied:

1. Linear Regression

- Simple baseline regression model
- Assumes linear relationship between features and output

2. Decision Tree Regression

- Non-linear model
- Splits data into decision nodes

3. Random Forest Regression

- Ensemble method
- Combines multiple decision trees
- Reduces overfitting

4. Support Vector Regression (SVR)

- Uses kernel-based approach (RBF kernel)
- Works well for complex relationships

5. Model Evaluation Results

Model	***R^2 Score***
Linear Regression	0.75 (approx)
Decision Tree Regressor	0.65 – 0.85
Random Forest Regressor	0.85 – 0.92
SVR (RBF Kernel)	0.70 – 0.85

6. Final Model Selection

Selected Model: Random Forest Regressor

- Highest and most stable R^2 score
- Handles non-linear relationships effectively
- Reduces overfitting compared to Decision Tree

7. Conclusion

In this project, multiple regression algorithms were applied to predict insurance charges. After evaluating performance metrics, the Random Forest Regressor was identified as the best-performing model.

This model provides reliable predictions and can assist insurance companies in estimating customer charges based on their profile.